Tutorial . Probability and Statistics (MCC505)
Monsoon Semester: 22-23

1. Suppose the average grade in a quiz is 70%. Give an upper bound of the proportion of students who scored atleat 90%.

2. Suppose a coin is flipped 2o times. Find a bound for the probability that head will come atleast 16 times. Take $P(H) = 0.2$

3. Suppose a fair coin is flipped 100 times. Find a bound of the probability that number of head will be atleast 60 or atmost 40.

4. Suppose someone gives you a coin and claims that this coin is biased; that it lands on heads only 48% of the time. You decide to test the coin for yourself. If you want to be 95% confident that this coin is indeed biased, how many times must you flip the coin? You can allow a precision error of 1%.

5. Let $X_1, X_2, \cdots, X_n$ be i.i.d. $U(0,1)$. Show that $Z_n \xrightarrow{P} c$ where c is a constant. Find c.

6. A computer has a random number generator that generates number between 0 and 1. Suppose we run it 100 times. Find $P(S_{100} \geq 52)$ applying CLT. Here $S_{100}$ denotes the sum of first 100 generated numbers.

7. By clinical study, it was found that people over age 50 had a mean cholesterol 54 and sd 17. Suppose a physician has 40 patients over age 50 and wants to determine the probability that the mean cholesterol of 40 patients are 60 or more. (Apply CLT).

8. Let $\{X_n\}$ be an i.i.d. sequence of standard normals. Let $Y_n = \frac{1}{n}\sum_{j=1}^{n} X_j^2$. Use the CLT to find n so that $P(Y_n \geq 1.01) = 5\%$.

9. (a). Let probability density of a random variable X be $f(x) = \begin{cases} \frac{5}{x^6} & \text{for } x \geq 1, \\ 0 & \text{otherwise.} \end{cases}$

What bound does Chebyshev's inequality give for the probability $P(X \geq 2.5)$? For what value of $\alpha$ can we say $P(X \geq \alpha) \leq 15\%$?

(b). Let $\{X_n\}$ be a sequence of random variables having common mean 5 and common variance 2. If the covariance between any two random variables in this sequence is negative, prove that Weak Law of Large Number holds for $\{X_n\}$.